Bayesian weather associations and daily time structure in multi-city dock-based bike share

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Abstract

Weather is one of the few drivers a bike-share operator cannot control. This project continues the midterm panel: a public daily dataset derived from the release behind Bean et al. [2021a], with 29 systems and 10,578 city-days. We estimate hierarchical Student-t regressions with city weather coefficients drawn from a common population. The association analysis keeps 25 continuously operating systems and the prediction evaluation keeps 23 covering all of 2017. Precipitation is the stable result: a $+10$ mm step from the city median is associated with a -20.21% change in the conditional median of $(1 + \text{usage})$ under M2 (95% interval $[-21.37, -18.99]$) and -19.98% under M0. The temperature association is positive over the full period under both specifications, but the interior peak moves with the specification, so we name no optimal temperature. M2 shows lower lag-one correlation in its innovation residuals and lower prediction scores in the development windows, but its block forecast on the pre-specified final window is worse than the archived model, with 90% coverage of 60.5%. The specifications differ in several structural components, so the contribution of AR(1) alone is not established.

1 Introduction

Bike-share operators plan rebalancing, pricing and maintenance around demand that moves with the weather. The comparative study of Bean et al. [2021a] fits a separate generalised additive model to each of forty cities and reports a thermal response that usually peaks near 27–28°C and a negative rainfall response. Fitting each city separately is a natural first step, and it leaves two questions open. First, with roughly one year of daily data per city, how much of the apparent city-to-city spread in weather sensitivity is signal rather than noise? Second, does treating days as exchangeable observations leave structure in the residuals that matters once the model is asked to predict?

We continue the midterm questions about weather associations, city variation and city backgrounds, using the same dataset. We also evaluate daily time structure as a predictive extension.

1. Once calendar effects and seasonal variation are controlled, what is the association between precipitation and daily usage, and how does it vary across cities and rainfall backgrounds?
2. What are the direction, shape and uncertainty of the temperature association, and how do they change across model specifications and city thermal backgrounds?

40th Conference on Neural Information Processing Systems (NeurIPS 2026).

3. What do calendar, trend and AR(1) (first-order autoregressive) residual structure change about diagnostics and prediction in the development windows, and does the final window support extending those results?

The associations are observational. We do not identify causal effects, we do not estimate climate adaptation, and because both prediction tasks condition on realised weather, we make no claim about operational forecasting that includes weather-forecast error.

2 Data and evaluation design

Panel and rosters. The frozen input is a daily panel derived from the Mendeley Data release [Bean et al., 2021b] (Version 1, CC BY 4.0) that accompanies Bean et al. [2021a]. Hourly hires inferred from station inventory snapshots are aggregated to local dates and paired with daily summaries of the Universal Thermal Climate Index (UTCI), a thermal-stress index that combines air temperature, humidity, wind and radiation and is not the same quantity as air temperature, together with daily precipitation from the Copernicus Climate Data Store [Copernicus Climate Change Service, 2020]. The panel holds 10,578 city-days for 29 systems with no duplicate city-date keys, no missing cells and no negative values. Usage is inferred from inventory changes, may include rebalancing moves, and is not a count of individual trips.

Three scopes are fixed by the frozen metadata rather than by any result, and Figure 3 in the appendix shows them city by city. The association analysis keeps the 25 systems with `continuous_system` equal to one, that is, systems that record hires on at least 99% of their analysis days; these contribute 9,120 city-days. The prediction evaluation keeps the 23 of those systems whose study period spans 2017-01-02 to 2017-12-31, which excludes Kaohsiung and Melbourne because they cover 2016-07-02 to 2017-06-30; the roster contributes 8,393 city-days. The prediction roster is therefore conditioned on full-period activity, and it is not re-selected using test-period usage.

Time splits. Every model is trained only on observations at or before a common cutoff, so a model predicting one city's June cannot learn from another city's November. We use two development folds and one pre-specified final fold, each with a 56-day test block: cutoff 2017-07-16 with test 2017-07-17 to 2017-09-10; cutoff 2017-09-10 with test 2017-09-11 to 2017-11-05; and cutoff 2017-11-05 with test 2017-11-06 to 2017-12-31. Training rows are 4,529, 5,817 and 7,105; every test block is $23 \times 56 = 1{,}288$ rows. The final block was included in the full-period midterm exploratory analysis, so its observations were already available before the final project. We report this pre-specified evaluation with that disclosure. All centring constants, scales and moderators are computed from training rows only.

Prediction tasks. The *block* task fixes a start date and predicts the next 56 days without reading any realised usage after the start, propagating the residual state forward from the posterior. The *one-step* task predicts day t from realised usage through day $t-1$, with the parameters fixed at the start of the window. Both use realised weather, so neither includes weather-forecast error, and the two tasks are not directly comparable because they condition on different information.

Metrics. We report the weighted interval score (WIS), the log-scale mean absolute error (MAE) and the 90% interval coverage, all on the $\log(1+\text{usage})$ scale and aggregated with equal city weights, so that no single large system dominates. The appendix gives the full definitions, the additional normalised error, and the four baselines. Because the multi-step intervals of the seasonal naive and AR-OLS baselines do not accumulate future innovation uncertainty, we use the baselines only for point-forecast comparison and do not base any model-superiority claim on their interval scores, densities or coverages.

3 Models and priors

All models share the response $y_{it} = \log(1 + U_{it})$ for city i on local date t, a Student-t likelihood with a shared degrees-of-freedom parameter ν and a city-specific scale σ_i, and the linear predictor

$$\mu_{it} = \alpha_i + \beta_i^{\text{temp}} T_{it} + \beta_i^{\text{sq}} Q_{it} + \beta_i^{\text{rain}} R_{it} + \text{calendar}_{it} + \text{trend}_{it}, \tag{1}$$

Table 1: Prior scales in the fitted models, on the $\log(1 + \text{usage})$ scale. "Population" is the standard deviation of the normal prior on the population mean; "between city" is the scale of the half-normal prior on the between-city standard deviation.

Term	Population	Between city
Intercept	$\mathcal{N}(7,\ 2)$	HalfNormal(2)
UTCI, per 10°C	$\mathcal{N}(0,\ 0.5)$	HalfNormal(0.5)
UTCI squared	$\mathcal{N}(0,\ 0.3)$	HalfNormal(0.3)
Precipitation	$\mathcal{N}(0,\ 0.3)$	HalfNormal(0.3)
Climate moderator, thermal	$\mathcal{N}(0,\ 0.2)$	—
Climate moderator, rainfall	$\mathcal{N}(0,\ 0.3)$	—
Weekend indicator	$\mathcal{N}(0,\ 0.4)$	HalfNormal(0.4)
Weekday contrasts, six free	$\mathcal{N}(0,\ 0.4)$	—
Annual harmonic pair	$\mathcal{N}(0,\ 0.8)$	HalfNormal(0.8)
City trend, per 100 days	$\mathcal{N}(0,\ 0.3)$	HalfNormal(0.3)
Log noise scale	$\mathcal{N}(-0.5,\ 0.8)$	HalfNormal(0.5)
AR persistence z_i	$\mathcal{N}(0,\ 0.5)$	HalfNormal(0.5)

where T_{it} is the daily mean UTCI minus the city's training-period mean, in units of 10°C; Q_{it} is T_{it}^2 centred within city on its training mean; and R_{it} is $\log(1 + \text{precipitation})$ minus the city training mean, divided by the pooled within-city training standard deviation.

Hierarchy. Each city coefficient is drawn from a normal population whose mean is a model parameter and whose between-city scale is a half-normal parameter estimated from the data [Gelman et al., 2013]. The city intercept uses a centred form, $\alpha_i \sim N(\alpha_{\text{pop}}, \tau_\alpha)$, and the city noise scale is centred in the same way; the weather coefficients keep the non-centred form $\beta_i = \beta_{\text{pop}} + \tau_\beta \tilde{\beta}_i$ with $\tilde{\beta}_i \sim N(0, 1)$. The two forms imply the same prior on the coefficients, and the difference is computational. This is partial pooling. The cities are neither fitted independently nor forced to share a single coefficient; the data decide how far each city's estimate is pulled toward the common mean, and how strong that pull is for a particular city depends on how much information its own observations carry relative to the between-city spread. We did not run a check that ranks cities by the amount of shrinkage, so we do not report which cities are shrunk most.

Specifications. M0 is the archived reference. Its temperature and precipitation population means shift with the city's thermal and rainfall background through a moderator term, its calendar effect is a weekend indicator, and its annual cycle is one sine/cosine pair. M1 replaces the weekend indicator with a shared weekday effect, parameterised through an orthonormal sum-to-zero basis with six free contrasts, and adds a shrunk city-specific linear trend per 100 days centred on the training-period mean day number. M2 adds a city-specific AR(1) residual process with Student-t innovations,

$$e_{it} = \phi_i e_{i,t-1} + \eta_{it}, \qquad \eta_{it} \sim t_\nu(0, \sigma_i), \qquad \phi_i = \tanh(z_i), \tag{2}$$

with a hierarchical prior on z_i. The first observation of every contiguous date segment uses a wider initial distribution than the stationary innovation scale, and the AR recursion restarts whenever the calendar gap exceeds one day. The stationarity of that initial distribution and its sensitivity to the choice were not verified in this project, so we describe it as a wide initial prior. Because σ_i in M2 is an innovation scale, it is not comparable with the residual scale of M0 or M1. M0 and M2 differ in the calendar effect, in the presence of a trend, and in the residual process at the same time, so the lower innovation-residual correlation reported in Section 4 is a property of the whole change and cannot be attributed to the AR(1) term on its own.

Priors. Table 1 lists the scales that the fitted models actually use, read from the model constructor. For the varying coefficients other than the noise scale, the constructor takes a single scale argument and uses it twice, once as the standard deviation of the population normal and once as the scale of the half-normal prior on the between-city standard deviation, so those rows have matching entries; the noise row does not. The intercept and weather scales are the archived M0 scales and are kept for comparability with the earlier results, and the group scales follow Gelman [2006]. The trend uses $\mathcal{N}(0, 0.3)$ with a HalfNormal(0.3) group scale. The degrees of freedom are $\nu = 2 + \text{Exponential}(0.1)$, which places prior mass on heavy tails without excluding near-normal ones.

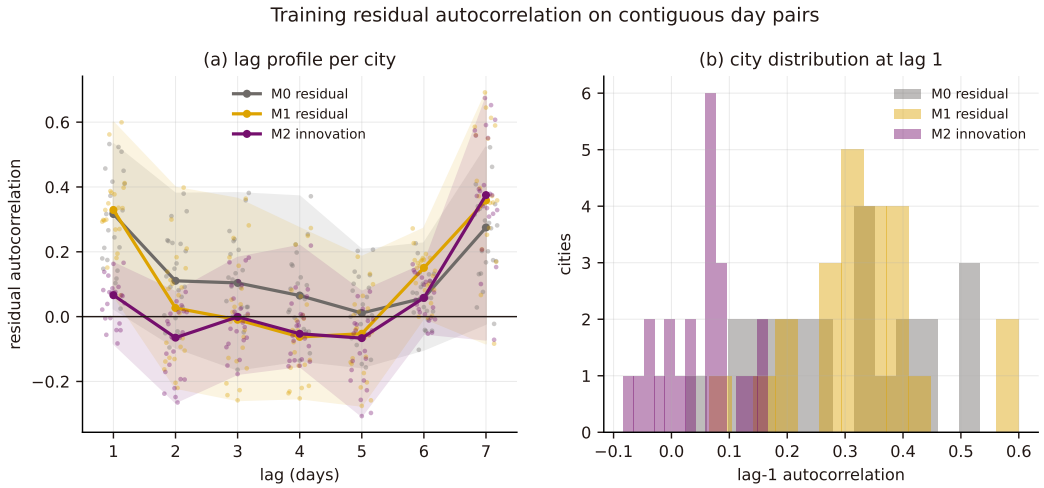


Figure 1: Training residual autocorrelation on contiguous day pairs, first development fold. Panel (a) lag profile per city with the median line; panel (b) the city distribution at lag one. M0 and M1 show fitted residuals; M2 shows innovation residuals $e_{it} - \phi_i e_{i,t-1}$.

Prior predictive check. A supplementary check on M2 for the first development fold generates the full mean function from the prior with the observed weather and calendar, recurses the AR state and simulates Student-t innovations. With 1,000 parameter draws it places 3.50% of the log response below zero, so the negative fraction is small while the upper tail is wide and the prior is broad rather than tight. The check was not run for M0, and we do not claim it was. The same likelihood permits negative log usage, and after the exponential transform the expected usage does not exist for finite ν; that bound applies to any point summary and is carried into the discussion.

4 Inference and diagnostics

Inference uses NUTS through PyMC [Abril-Pla et al., 2023] and nutpie [Seyboldt, 2024], with four chains of 3,000 warmup and 3,000 retained draws, seed 27, `target_accept` 0.92 and maximum tree depth 12. A run enters the summary only if every parameter satisfies $\hat{R} \leq 1.01$ [Vehtari et al., 2021, Gelman and Rubin, 1992], bulk and tail effective sample size of at least 400, no divergences, no tree-depth saturation, and a Monte Carlo standard error below 5% of the posterior standard deviation [Betancourt, 2017]. Across the 11 core fits the worst rank-normalised $\hat{R}$ is 1.00894, the smallest bulk ESS is 426.7 and the smallest tail ESS is 668.9, with no divergences and no tree-depth saturations. These numbers establish that the numerical summaries are reliable at the stated precision. They do not by themselves establish that the model fits the data or that its predictions are useful, and the rest of the report treats those as separate questions. The appendix records the three reparameterisations adopted for computational reasons and the fitted time-structure parameters.

Residual time dependence. The archived model's training residuals are not white. On the first development fold the median lag-one autocorrelation across the 23 cities is 0.317 under M0 and 0.329 under M1, because calendar terms cannot absorb persistent day-to-day dependence, while M2's innovation residuals fall to 0.066 on the first development fold, -0.003 on the second and -0.036 on the final window. Figure 1 shows the lag profile and the city distribution at lag one. M2 therefore shows lower lag-one correlation in its innovation residuals, and the prediction results in Section 6 qualify that comparison. Because the specifications differ in several structural components, these comparisons do not establish the contribution of AR(1) alone.

Training predictive checks. Because the AR(1) model conditions each day on the previous observed residual, the training-period 90% coverage under M2 is a conditional one-step check given past observations and not a joint replication of a whole time series. The appendix reports the coverage values and the corresponding figure under that label.

5 Weather associations

We report the change in the conditional median of $(1 + \text{usage})$ for a $+10$ mm step from the city's median daily precipitation, and for a $+10°$C step from the city's training-period mean UTCI. On the $\log(1 + \text{usage})$ scale a step of $\Delta\mu$ multiplies $(1 + \text{usage})$ by $\exp(\Delta\mu)$, so the percentage is $100(\exp(\Delta\mu) - 1)$; it is computed for each posterior draw and each city and then averaged with equal city weights. The quantity is a relative change in the modelled location of $1 + \text{usage}$, and the expected change in raw usage cannot be recovered from a Student-t log-scale model by simple back-transformation. Table 4 in the appendix reports both full-period fits side by side.

Precipitation. Precipitation is the association this project would defend. In the 25-city full-period fit the city-equal average is -20.21% (95% interval $[-21.37, -18.99]$) under M2 and -19.98% ($[-21.07, -18.89]$) under M0, and the posterior probability that the population coefficient is negative is 1.000 in both fits. Replacing the fixed $+10$ mm step with an observation-supported contrast, from the city median to the 90th percentile of daily precipitation, gives -16.99% ($[-18.02, -15.95]$) under M2 and -16.40% ($[-17.28, -15.53]$) under M0, so neither the sign nor the order of magnitude depends on the contrast definition.

The association is negative in every one of the 25 cities, with city medians spanning -36.18% to -10.05% under M2 and -33.84% to -8.03% under M0, and the smallest city-level posterior probability of a negative coefficient is 0.984 under M2 and 0.998 under M0. Figure 2 shows the city estimates, each taken from its own saved fit. The spread across cities is substantial, but the city intervals overlap and we have not tested the medians against one another, so we do not read the ordering as evidence that two particular cities differ.

Temperature. Over the full period the city-equal $+10°$C effect is positive under both specifications: 5.22% ($[3.67, 6.76]$) under M0 and 6.16% ($[4.12, 8.32]$) under M2, on the same 25 cities and the same window. The location of the response peak remains uncertain across specifications.

That uncertainty is visible in how often a concave curve peaks inside the observed support: 15 of 25 cities under M0 and 5 of 25 under M2 have a posterior probability above 0.95 of an interior maximum. The two counts come from different fits and must not be combined, and the peak location is always conditional on an interior peak existing, so we report the existence probability next to it. In both fits exactly 1 city has a $+10°$C contrast that leaves its observed range, namely Kaohsiung, whose study period is the warmest in the panel. Figure 4 in the appendix shows the curve for three cities with their observed support, and Figure 5 repeats all 25 curves inside the observed range. One possible source of the variation across specifications is that every model already carries an annual harmonic pair fitted to the same city means, so the temperature coefficient is estimated alongside a seasonal control that shares much of its variation; we did not run a decomposition that would confirm that overlap as the cause.

City-background associations. The midterm's third question concerns whether city backgrounds are associated with differences in weather coefficients. For weather term k, the hierarchy has the form $\beta_i^k = \bar{\beta}^k + \gamma_k Z_i^k + \tau_k u_i^k$, with $u_i^k \sim N(0, 1)$. For the two thermal terms, Z_i^k is the city's mean UTCI standardised across cities; for rain it is the standardised $\log(1 + \text{mean daily precipitation})$. Each city receives equal weight in this standardisation, and prediction fits compute the backgrounds from training observations. Table 2 reports the saved full-period estimates. In both specifications, the intervals for the rainfall and thermal linear moderators include zero. The thermal quadratic moderator is negative in both, associating a warmer study-period background with a lower quadratic coefficient. This is an association between fitted coefficients and city backgrounds; each city has about one year of data, so its interpretation is limited to that observation period.

6 Prediction and evaluation

Interval forecasts. Table 3 is the primary prediction result. On the two development folds M2 has the lowest block WIS, 0.1265 and 0.1577 against 0.1338 and 0.2046 for M0 and 0.1320 and 0.1893 for M1, and its mean development WIS is 0.1421 against 0.1606 for M1 and 0.1692 for M0. The pre-specified selection rule, applied to the development folds only, therefore selects M2, and the same specification is evaluated on the final window. The paired seven-day block comparison of M2 against

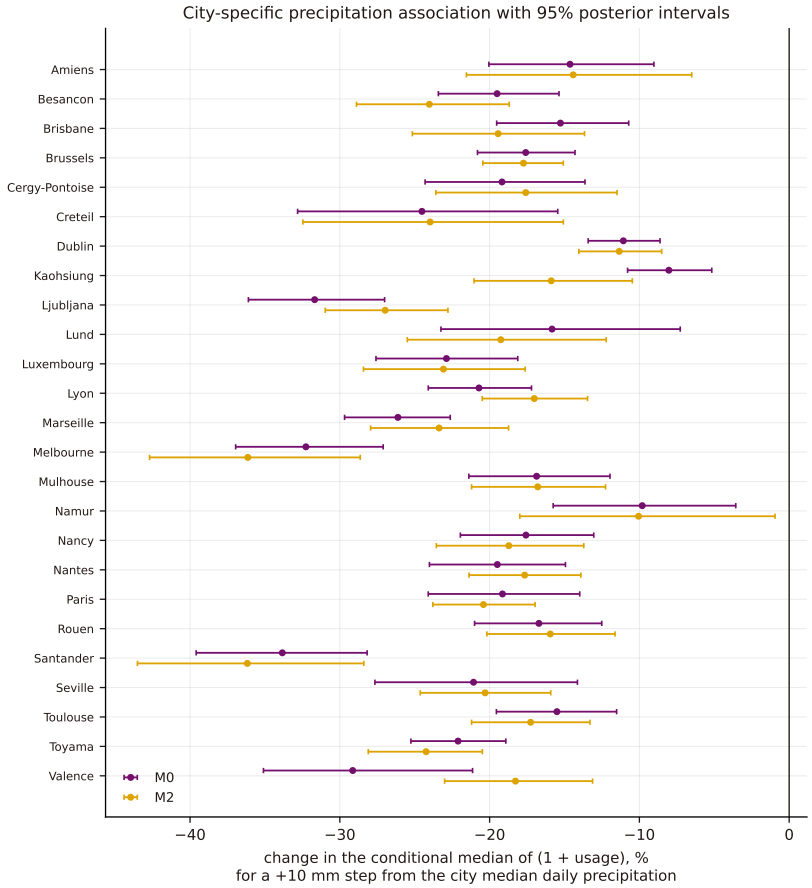


Figure 2: City-specific precipitation association in the 25-city full-period fits, shown separately for M0 and M2. Points are posterior medians and bars are 95% posterior intervals for the percentage change in the conditional median of $(1 + \text{usage})$ for a $+10$ mm step from the city median.

Table 2: City-background moderators in the 25-city full-period fits. Entries are posterior medians and 95% intervals. Each moderator is the change in its weather coefficient per one standard deviation of the corresponding city background; entries are coefficients on the log-response scale.

Weather coefficient	M0	M2
Rainfall	-0.0031 $[-0.0210,\ 0.0136]$	-0.0147 $[-0.0332,\ 0.0021]$
UTCI linear	-0.0320 $[-0.0772,\ 0.0163]$	-0.0179 $[-0.0510,\ 0.0165]$
UTCI quadratic	-0.0282 $[-0.0504,\ -0.0056]$	-0.0150 $[-0.0250,\ -0.0051]$

M0 gives $[-0.0153, 0.0011]$ on the first development fold and $[-0.0675, -0.0231]$ on the second; the first interval contains zero, so only the second window shows a difference separated from zero.

The final window. The development-window advantage does not carry over. On the final block M2 scores 0.3394 against 0.1957 for M0, and its 90% coverage is 60.5% against 75.5%. The one-step task shows the same pattern with 0.2085 against 0.1957 for M0 and coverage of 72.0% against 75.7%. M2 has wider mean intervals than M0, at 1.095 log units against 0.844, but lower coverage. These summaries indicate a mismatch between the predictive distributions and the observations. Location error is a possible contributor, but the mechanism has not been established. We describe this as a limitation of the specification on this window rather than as evidence about the wider population of bike-share systems, because a single final window is one piece of evidence. The gap is not attributed to any single term: we did not fit an M2 variant without the city trend, so the attribution remains unverified, and the appendix records the fitted trend values and their time scale as an observation. The

Table 3: Interval forecasts for the fitted models. WIS is city-equal on the $\log(1 + \text{usage})$ scale and lower is better; coverage is the city-equal 90% interval coverage. Block and one-step condition on different information and are not comparable.

	City-equal WIS			90% coverage (%)		
Model and task	dev 1	dev 2	final	dev 1	dev 2	final
M0, block	0.1338	0.2046	0.1957	76.6	53.8	75.5
M1, block	0.1320	0.1893	0.3690	85.1	70.8	54.9
M2, block	0.1265	0.1577	0.3394	88.5	80.1	60.5
M2, one-step	0.1084	0.1238	0.2085	90.3	87.2	72.0

mean log predictive density also carries visible Monte Carlo noise, so we treat the WIS comparison as the more reliable summary; the appendix gives the re-run values.

Point forecasts against the baselines. The baselines are compared on log-scale MAE only. On the final block the log-scale MAE is 0.3207 for M0, 0.4252 for the seasonal naive and 0.5547 for M2. On the one-step task it is 0.3197 for the city AR-OLS, 0.3207 for M0 and 0.3601 for M2. All values are city-equal on the $\log(1 + \text{usage})$ scale and lower MAE is better; the complete table, including the city OLS, the city constant and the remaining one-step values, is Table 5 in the appendix.

7 Discussion and limitations

The precipitation association answers the first question. It is negative in every city, it is stable across the two full-period specifications, across the fixed-step and observation-supported contrasts, and across the three model specifications on the development fold, and it is bounded away from zero in the city-equal summary. The city estimates differ in size, and the intervals overlap, so the hierarchy describes a shared signal with city-level variation rather than a confirmed ranking.

The temperature association is positive under both full-period specifications, and the peak location is not identified well enough to name a single optimal temperature, because the estimate moves with the specification and with the contrast definition. Both statements are conditional associations from observational data.

The city-background estimates complete the midterm question about between-city variation. The rainfall and thermal linear moderators remain uncertain, while the thermal quadratic moderator is negative under both full-period specifications. Study-period backgrounds therefore provide limited evidence about the form of city variation, with the clearest association in the quadratic term.

The time-structure result answers the third question in two parts. M2 shows lower lag-one correlation in its innovation residuals and lower prediction scores in the development windows, but the comparison does not establish the contribution of AR(1) alone, because the specifications differ in several structural components. That development-window advantage does not survive into the final window, where the block forecast is worse than the archived specification and coverage falls to 60.5%. We did not retune after seeing the final window, and we do not present the richer specification as a general improvement. Sampling reliability, residual diagnostics and predictive performance are three separate claims, and only the first is settled by the convergence results in Section 4.

The reported prior-sensitivity analysis compares weather scales of 0.5, 1 and 2 times the baseline for M2 on the first development fold. The full 29-city roster, a Gaussian likelihood and removal of seasonal terms are unassessed extensions. The reported findings are conditional on the continuous-system roster, Student-t observations and the annual harmonic control used here.

The limitations are short and bound what the evidence supports. The study is observational and cannot identify causal effects or climate adaptation. Each city has about one year of data, so the study-period mean is an observation-period background rather than a long-run climate normal, and the prediction roster is conditioned on full-period activity, so the evaluation describes continuously operating systems. Usage is inferred from inventory changes and may include rebalancing moves. Both prediction tasks condition on realised weather and therefore say nothing about a forecast that has to predict the weather first, and the final block was inspected before this analysis, which we note wherever its numbers appear. The Student-t log-scale likelihood permits negative log usage

and the expected usage does not exist after the exponential transform, so we report medians and predictive quantiles only. Finally, the three models differ in several structural respects at once, so the comparison identifies the value of the package rather than the value of any single term.

Contribution statement

XUE Yantong searched for data, selected the topic, designed the experiments, developed code, ran the experiments, conducted the main statistical analysis, and prepared the report and presentation. WANG Ziyu and ZHAO Jiashen contributed to data processing, code development and checking, and verification of the report and slides.

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A Appendix

Panel by city. Figure 3 shows the three rosters, the system sizes and the observation windows. Every city except Kaohsiung and Melbourne starts on 2017-01-01 or 2017-01-02 and ends on 2017-12-31, so 21 of the 23 prediction cities contribute 365 days and Brisbane and Toyama contribute 364. Kaohsiung has 363 recorded days and one missing date, 2016-09-27, the only gap inside a study period anywhere in the panel; it affects only the full-period fit for the 25-city roster, because the AR recursion restarts after any calendar gap longer than one day. Usage is recorded as a half-integer on 4,936 city-days.

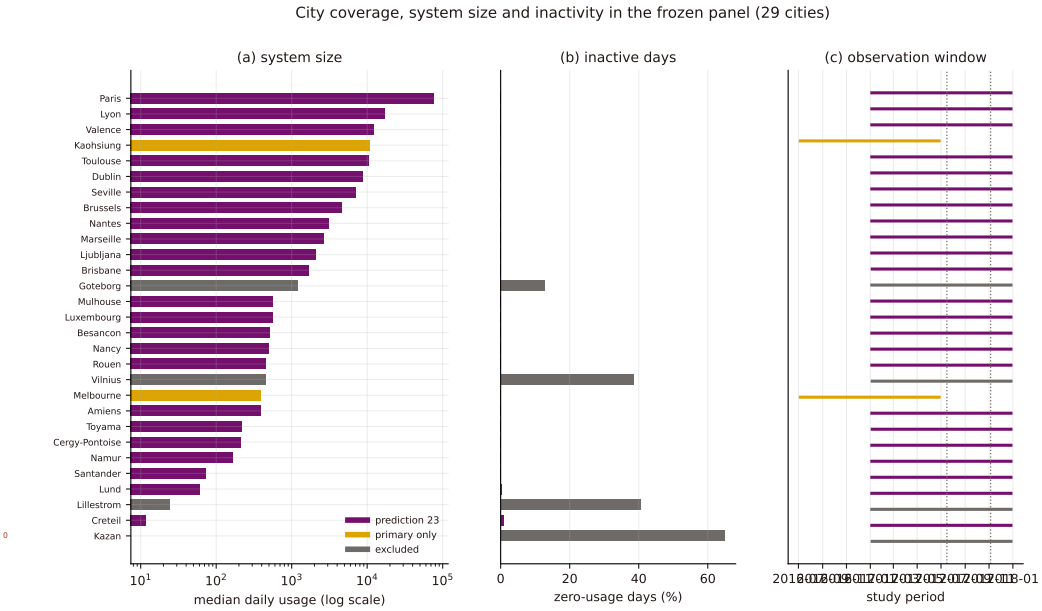


Figure 3: The frozen panel by city, ordered by median daily usage. Panel (a) system size on a log scale, panel (b) share of zero-usage days, panel (c) observation window with the development and final cutoffs marked. Colours separate the 23 prediction cities, the two primary-only cities and the four excluded systems.

Metric definitions. The WIS uses the fixed 50/80/90/95% central intervals, following Bracher et al. [2021] and the scoring-rule background in Gneiting and Raftery [2007]. The mean log predictive density is the average log density at the realised value, the log-scale MAE is the average absolute error in $\log(1 + \text{usage})$, and the 90% interval coverage is the share of realised values inside the central 90% interval. The normalised error divides each city's log-scale absolute error by that city's training-period seven-day seasonal naive error. The four baselines are a city constant, a seven-day seasonal naive forecast, a city OLS with parameter uncertainty, and a city AR-OLS.

Implementation details of the fitted models. The stored configuration carries a `beta_trend_city_sd` entry of 0.2. The constructor does not read it: it passes `beta_trend_population_sd` $= 0.3$ to the varying-coefficient helper, which uses that single value for both the population mean and the between-city scale, so the fitted trend prior is $\mathcal{N}(0, 0.3)$ with a HalfNormal(0.3) group scale. We report the fitted value and leave the configuration file unchanged so that the saved run records stay as they were produced. The first observation of every contiguous date segment uses a Student-t distribution with scale $\sigma_i \sqrt{\nu/((\nu-2)(1-\phi_i^2))}$, which is wider than the stationary innovation scale.

Prior predictive check. The saved prior predictive draw for M2 on the first development fold gives an intercept-only negative-usage fraction of 1.15%, which describes the city intercepts alone. The supplementary full check stored at `logs/review/followup_checks.json`, with its implementation at `logs/review/audit_followup.py`, uses 1,000 parameter draws and places 3.50% of the

log response below zero, with log-response quantiles $[-5.90,\ 19.46]$ against an observed range of 0 to 11.65. The corresponding full check was not run for M0.

Reparameterisations. Three components were reparameterised for computational reasons without changing the target distribution. The city intercept hierarchy and the city noise scale are written in centred form, and the shared weekday effect uses an orthonormal sum-to-zero basis with six free contrasts in place of a centred seven-vector that carried a direction the likelihood does not identify. Each change is a reparameterisation of the same prior and likelihood, and each was adopted after the original form failed the acceptance rule above.

Fitted time-structure parameters. In the M2 full-period fit the persistence parameter has a median of 0.498 with a 95% interval of $[0.147, 0.912]$ across cities and posterior draws, so the residual state carries information for roughly one to two days. The degrees of freedom are 2.76 $[2.55, 3.01]$, which keeps the innovation distribution heavy-tailed and places the fitted model away from the Gaussian limit. Under M0 the training-period 90% predictive coverage is 90.1% and under M2 it is 90.8%; Figure 10 shows the M2 check by city.

Full-period association table. Table 4 reports both full-period fits side by side, including the observation-supported contrast and the count of cities with an interior temperature peak.

Table 4: Weather associations in the 25-city full-period fits. Entries are city-equal averages of the percentage change in the conditional median of $(1 + \text{usage})$, with 95% posterior intervals. M0 and M2 are separate fitted specifications, reported from their own saved results.

Contrast	M0	M2
Precipitation: city median $+10$ mm	-19.98	-20.21
	$[-21.07, -18.89]$	$[-21.37, -18.99]$
Precipitation: median to 90th percentile	-16.40	-16.99
	$[-17.28, -15.53]$	$[-18.02, -15.95]$
UTCI: city mean $+10°$C	5.22	6.16
	[3.67, 6.76]	[4.12, 8.32]
UTCI: 25th to 75th percentile	19.89	14.04
	[17.44, 22.47]	[11.11, 17.27]
Cities with interior peak, $P > 0.95$	15 of 25	5 of 25
Cities leaving the observed range at $+10°$C	1 of 25	1 of 25

Development-fold temperature comparison. On the first development fold, which is a 23-city training window with a 56-day test block, the same $+10°$C quantity is 11.35% ($[9.72, 12.99]$) under M0, 10.07% ($[8.25, 12.01]$) under M1 and 7.49% ($[5.47, 9.65]$) under M2, while the precipitation association is almost unchanged across the same three fits at -18.17%, -17.94% and -18.18%. The observation-supported temperature contrast over the full period gives 19.89% ($[17.44, 22.47]$) under M0 against 14.04% ($[11.11, 17.27]$) under M2, and across the panel the observed daily mean UTCI runs from -37.8 to 34.3 degrees. The three development-fold values come from a different window and a different roster from the full-period values, and the two groups are not comparable with each other.

Point-forecast table. Table 5 reports the full final-window comparison, including the city OLS and the city constant, and Table 6 reports the same comparison on the two development folds.

Predictive integration stability. Re-running the same predictive integration for M2 with 800 posterior draws instead of 400 and a different seed gives a development-fold WIS of 0.12653 against the saved 0.12649 and a final-fold WIS of 0.33731 against the saved 0.33942, with a maximum absolute per-day density change of 0.126 and 0.363. The recomputed WIS values are close to the saved values, and the main model-comparison conclusion is unchanged; the results are not identical at the listed decimal places, and the per-day densities carry the larger Monte Carlo variation.

Fitted city trend. The city trend coefficients of the M2 final-window fit are positive in all 23 prediction cities, with city medians from 0.06 to 0.39 per 100 days and a city median of 0.17. The final test window lies 1.55 to 2.10 trend units beyond the training centre, so for a median city the

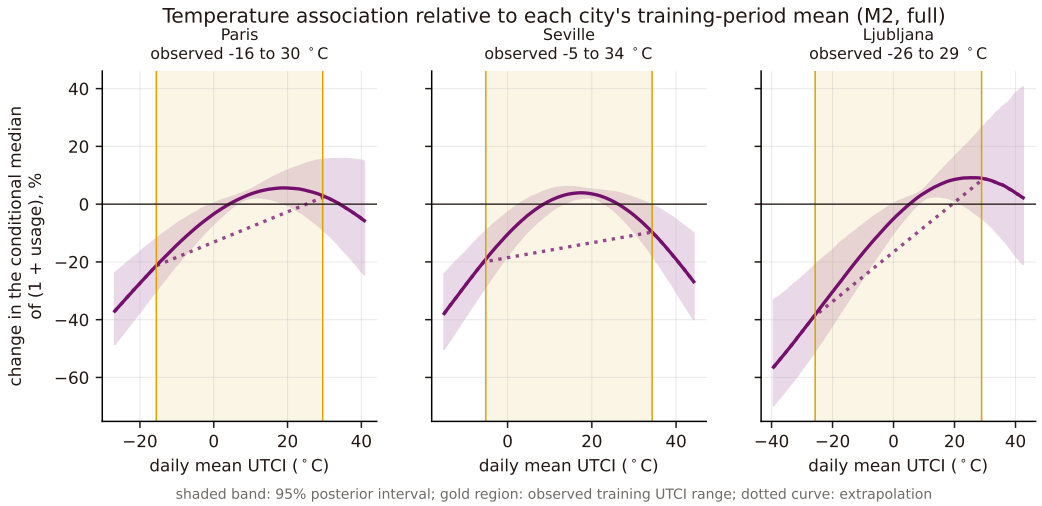


Figure 4: Temperature association relative to each city's training-period mean in the 25-city full-period M2 fit. Shaded bands are 95% posterior intervals; the gold region is the observed training UTCI range; dotted segments are extrapolation beyond that range.

Table 5: Point-forecast error on the final window, in log-scale MAE on $\log(1 + \text{usage})$, city-equal and lower is better. M0 and M2 come from the independent recomputation of their saved daily predictions; the four baselines come from a supplementary check completed before this write-up.

Method	Block MAE	One-step MAE
M0	0.3207	0.3207
Seasonal naive, 7-day lag	0.4252	0.3421
City AR-OLS	0.5192	0.3197
City OLS, parameter uncertainty	0.5208	0.5208
M2	0.5547	0.3601
City constant	0.5551	0.5551

trend term adds roughly 0.26 to 0.36 log units across the whole window. This is consistent with the trend contributing to a location error in the final window, but we did not fit an M2 variant without the trend, so the mechanism is not established.

Additional figures. Figure 6 shows the scores behind Table 3 across folds and tasks. Figure 7 shows the final-window block forecasts for three cities chosen before the plot was drawn: Paris, Lund, and the city with the median WIS. Figure 8 gives the convergence diagnostics across all parameters, Figure 9 the weather prior-scale check on the first development fold, and Figure 10 the training conditional predictive check.

Software and reproduction. Models were fitted with PyMC 6.3.2 [Abril-Pla et al., 2023] and nutpie 0.16.11 [Seyboldt, 2024] on PyTensor 3.3.2. Posterior summaries, rank-normalised $\hat{R}$ and effective sample sizes use ArviZ [Kumar et al., 2019]. Numerics use NumPy [Harris et al., 2020], pandas [McKinney, 2010] and SciPy. Figures use Matplotlib [Hunter, 2007]. The exact versions of every run are recorded in its `config.json`. The working directory contains the configuration, the source, the tests, the run outputs, the tables and the figures, together with a cleaned copy of the daily panel at `data/bikeshare_weather_daily.csv`, byte-identical to the frozen input used for every saved run and checked by SHA-256 before each fit. The pipeline is driven by `src/run.py` subcommands and described in `README.md`; the test suite asserts that the splits respect the cutoffs, that training standardisation does not depend on test rows, that block forecasts cannot read realised test usage, and that the AR recursion reduces to independent innovations when $\phi = 0$.

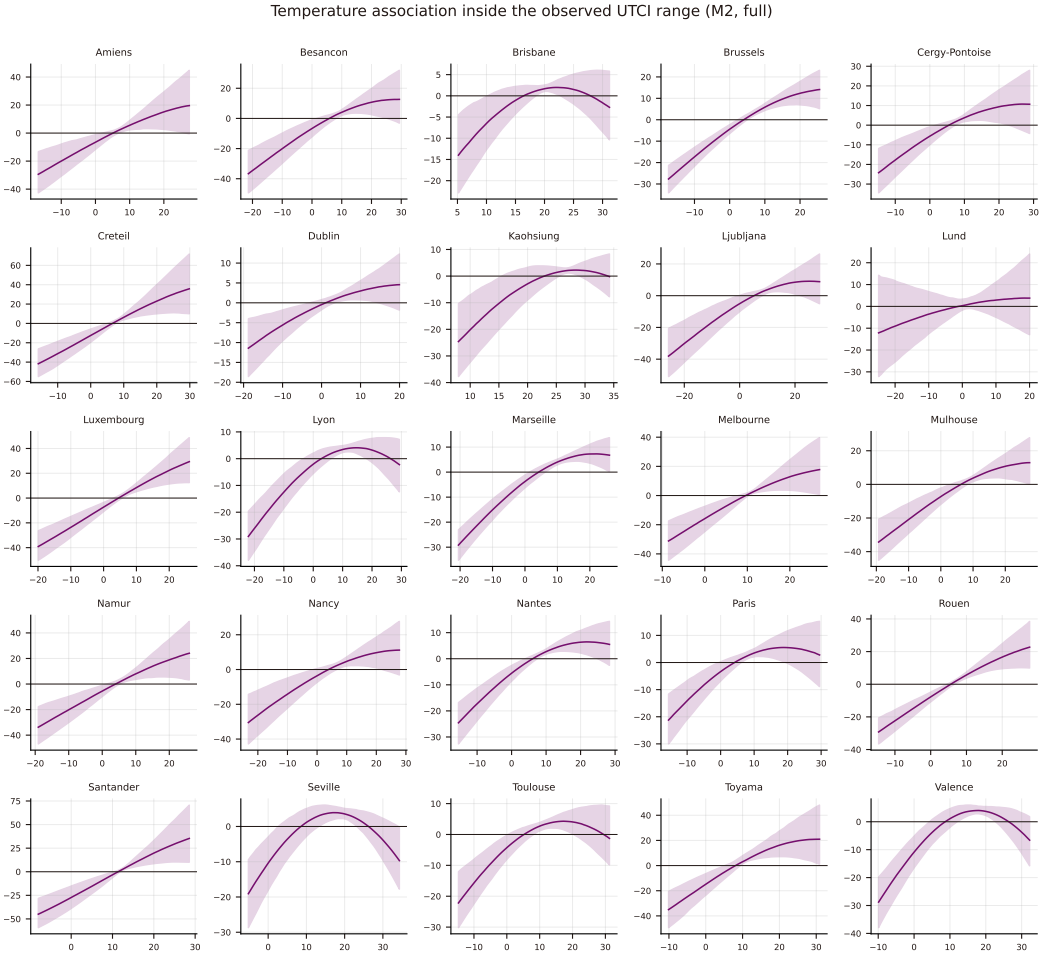


Figure 5: Temperature association inside the observed UTCI range for all 25 cities in the M2 full-period fit.

Table 6: Point-forecast error in log-scale MAE on $\log(1 + \text{usage})$, city-equal, lower is better, recomputed from the saved daily predictions.

Fold	Method	Block MAE	One-step MAE
dev 1	City constant	0.3757	0.3757
dev 1	Seasonal naive, 7-day lag	0.2602	0.2298
dev 1	City OLS, parameter uncertainty	0.3314	0.3314
dev 1	City AR-OLS	0.3309	0.2553
dev 1	M0	0.2571	0.2571
dev 1	M2	0.2503	0.2104
dev 2	City constant	0.3665	0.3665
dev 2	Seasonal naive, 7-day lag	0.2664	0.2405
dev 2	City OLS, parameter uncertainty	0.3131	0.3131
dev 2	City AR-OLS	0.3110	0.2278
dev 2	M0	0.3701	0.3701
dev 2	M2	0.3239	0.2540

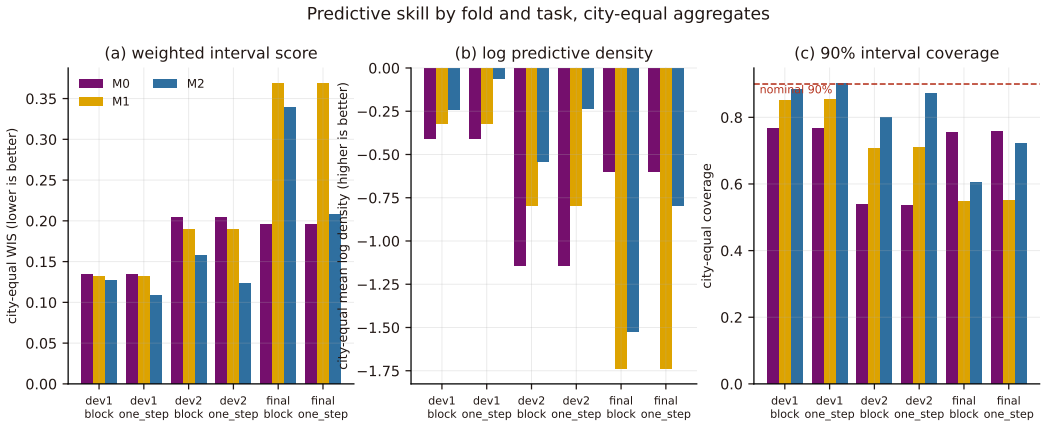


Figure 6: Predictive skill by fold and task. Panel (a) city-equal WIS, panel (b) city-equal mean log predictive density, panel (c) city-equal 90% coverage with the nominal level marked.

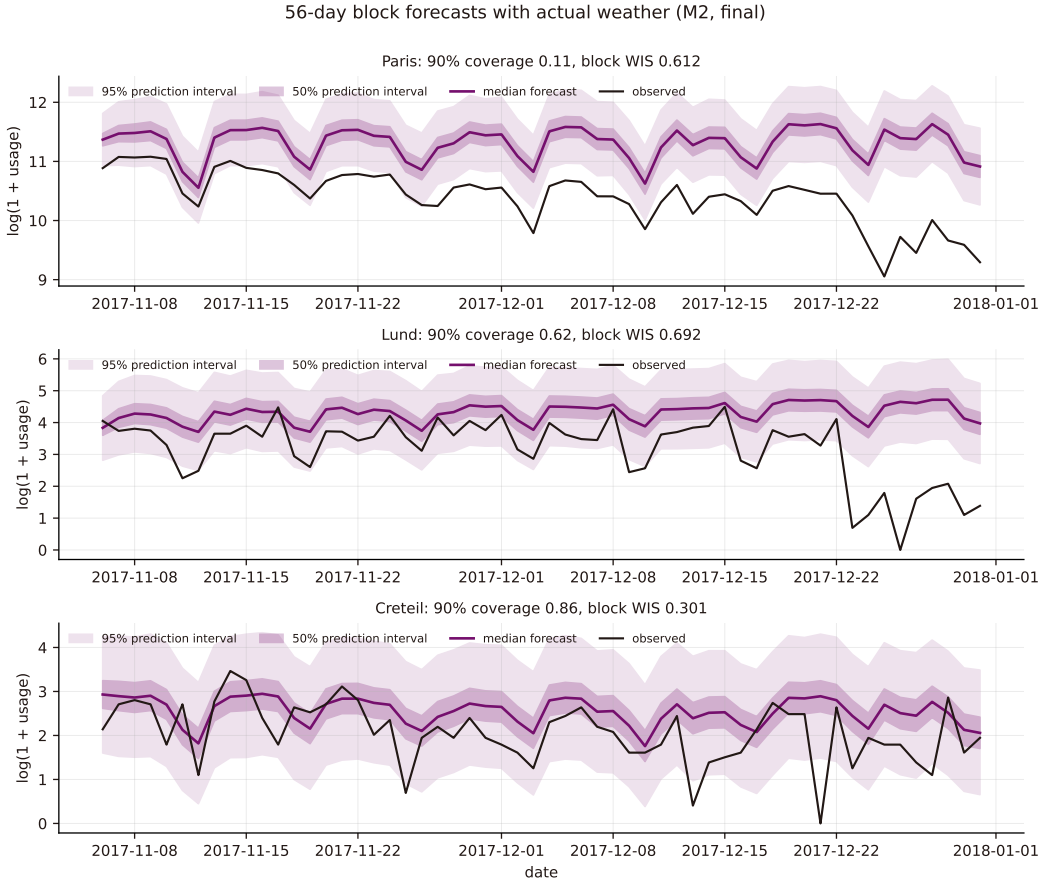


Figure 7: 56-day block forecasts with realised weather on the final window for Paris, Lund and the median-WIS city, under M2. Bands are 50% and 95% prediction intervals; the black line is the observed series.

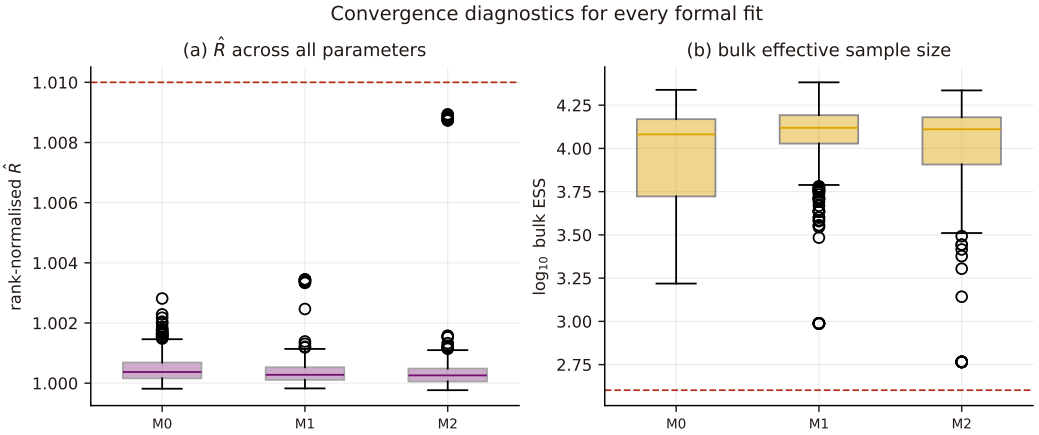


Figure 8: Convergence diagnostics for the M0, M1 and M2 fits on the first development fold, across all parameters.

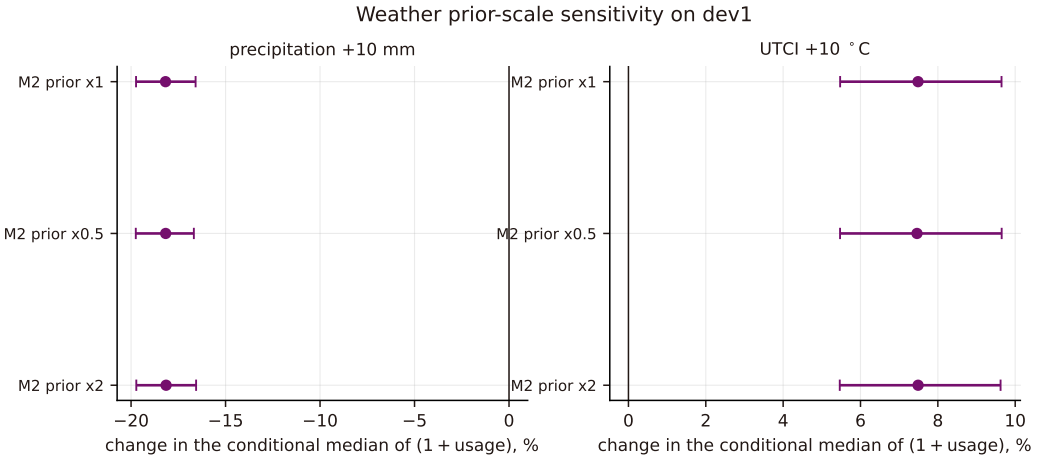


Figure 9: Weather prior-scale sensitivity for M2 on the first development fold. The rain association is -18.18% at the baseline scale, -18.17% at half scale and -18.15% at double scale; the temperature association is 7.49%, 7.46% and 7.49%.

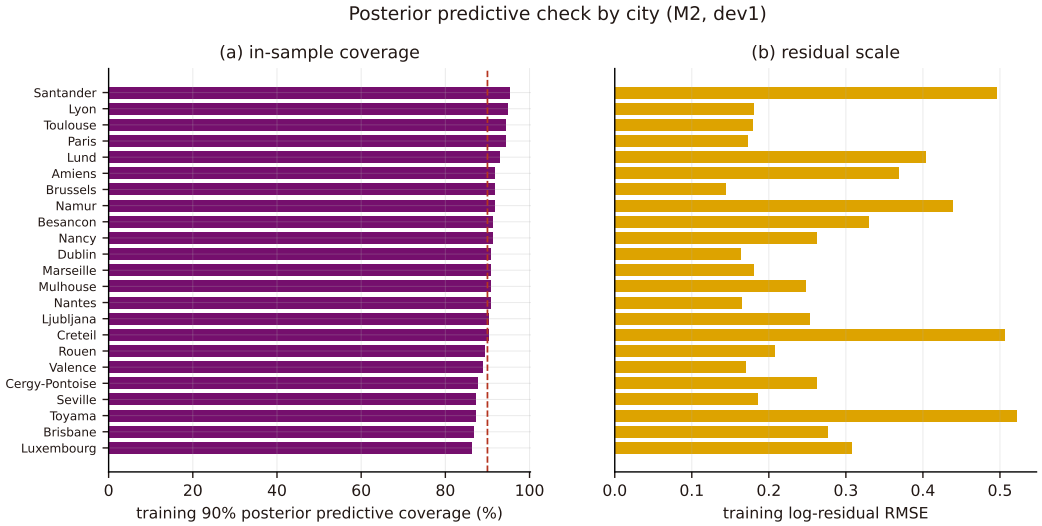


Figure 10: Training conditional predictive check by city for M2 on the first development fold. Each day is conditioned on the previous observed residual, so this is not a joint replication of a whole time series.